

Miqurios Platform Architecture

Engineering a signal from the noise.

This document summarizes the architecture of Miqurios and outlines how its design philosophy enables performance, adaptability, and long-term scalability.

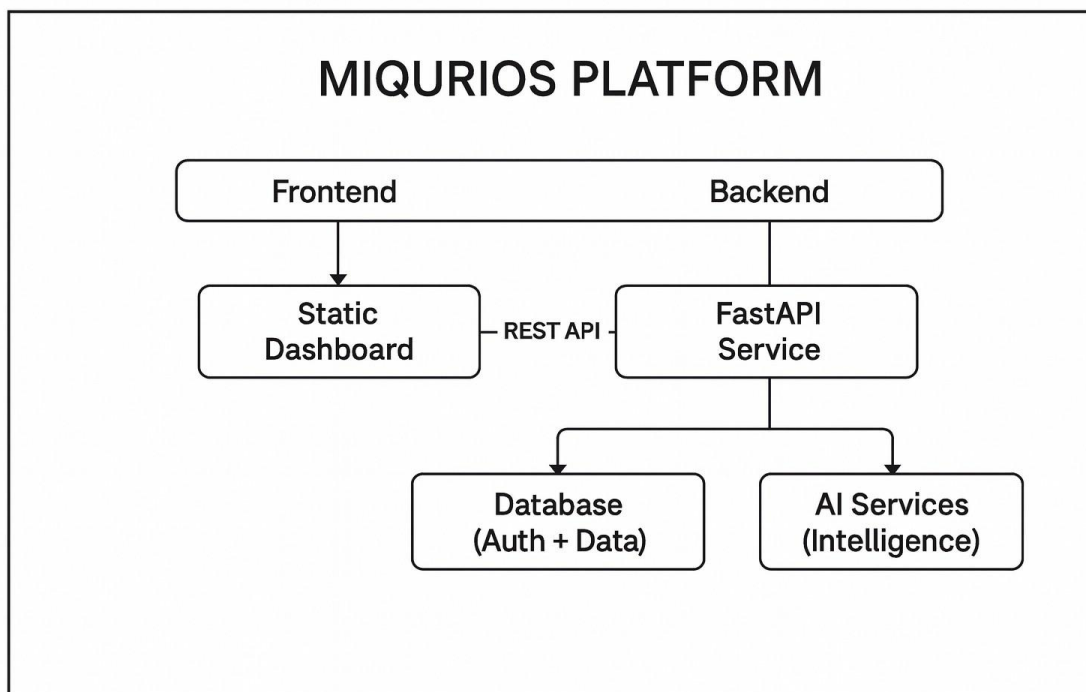
Overview

Content overload defines the modern world. Miqurios cuts through it with precision AI analysis, transforming siloed content into structured insight in seconds. This is not another summarizer or bookmarking tool. It is an intelligence layer for how thinking individuals consume information.

This overview presents the technical foundation and design philosophy behind Miqurios.

System Architecture

Miqurios is built on three principles: modularity, velocity, and intelligence.



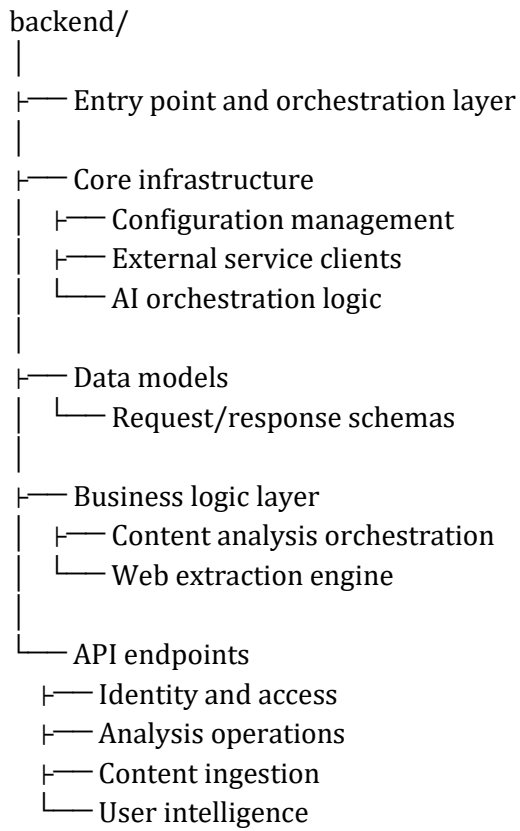
Stack

- Frontend: Static web application, globally distributed via CDN
- Backend: Python/FastAPI microservice architecture
- Intelligence: AI integration with custom orchestration layer

- Data: PostgreSQL with JWT authentication
- Infrastructure: Git-driven continuous deployment

Backend Architecture

The backend follows a layered microservice pattern designed for clarity and independent scalability.

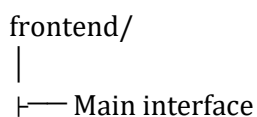


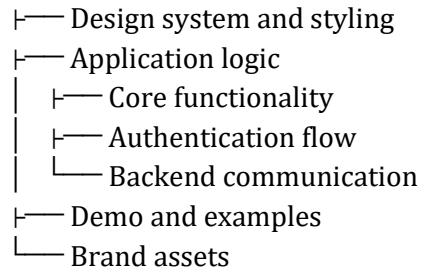
Why this structure matters

- New features deploy in hours, not weeks
- Issues are isolated by service boundary
- Multiple developers work without merge conflicts
- Services scale independently based on load

This is enterprise-grade architecture built with startup velocity.

Frontend

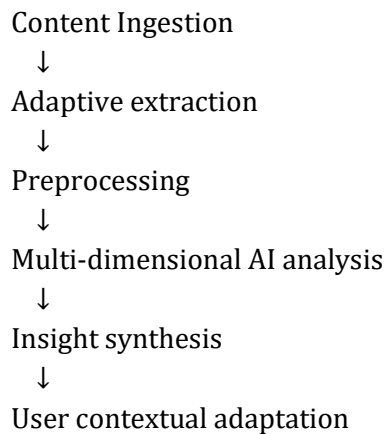




Design philosophy

- Minimal dependencies, maximum performance
- Global CDN distribution for sub-second load times
- Clean separation between UI and business logic

Intelligence Pipeline



Core approach

- Custom AI orchestration
- Adaptive extraction handles real-world web complexity
- Stateless design enables infinite horizontal scaling
- Type-safe contracts eliminate entire classes of bugs

Why This Architecture Wins

Velocity

Git push → Production in seconds. Zero manual deployment steps.

Resilience

Independent deployment of frontend, backend, and data layer. Failures stay contained.

Intelligence

AI is architectural, not supplemental. Context flows through every layer.

Clarity

Single-purpose modules. Explicit boundaries. Fast debugging and feature development.

Scale-Ready

Microservice extraction prepared. 10 users to 10 million without architectural rewrites.

Strategic Advantage

Miqurios learns from interaction itself. Each analysis session strengthens contextual understanding. The system becomes more adaptive and valuable over time, creating compounding intelligence that transforms a product into a platform.

Technical Differentiation

Typical MVP

Miqurios

Monolithic codebase

Layered microservice architecture

Manual deployments

Automated CI/CD from Git

Generic AI integration

Adaptive AI orchestration layer

Fragile web scraping

Adaptive extraction engine

Security as afterthought

JWT + OAuth from day one

Scaling = full rewrite

Scaling = configuration adjustment